☆: Yet unproved.

「Summary

$$\text{Maximal} \underset{\text{I.D}}{\overset{\text{Bézout}}{\Longleftrightarrow}} \text{Prime} \underset{\text{I.D}}{\overset{\text{Bezout}}{\Longleftrightarrow}} \text{Irreducible}.$$

$$\text{Maximal} \underset{\not\Leftarrow\ (x \in \mathbb{Z}[x,y],\ 2 \in \mathbb{Z}[x])}{\Rightarrow} \text{Prime} \underset{\not\Leftarrow\ (2 \in \mathbb{Z}[\sqrt{-5}])}{\Rightarrow} \text{Irreducible}.$$

Noether domain ⇒ Atomic domain.

If irr = prime.

Bézout domain

⇑

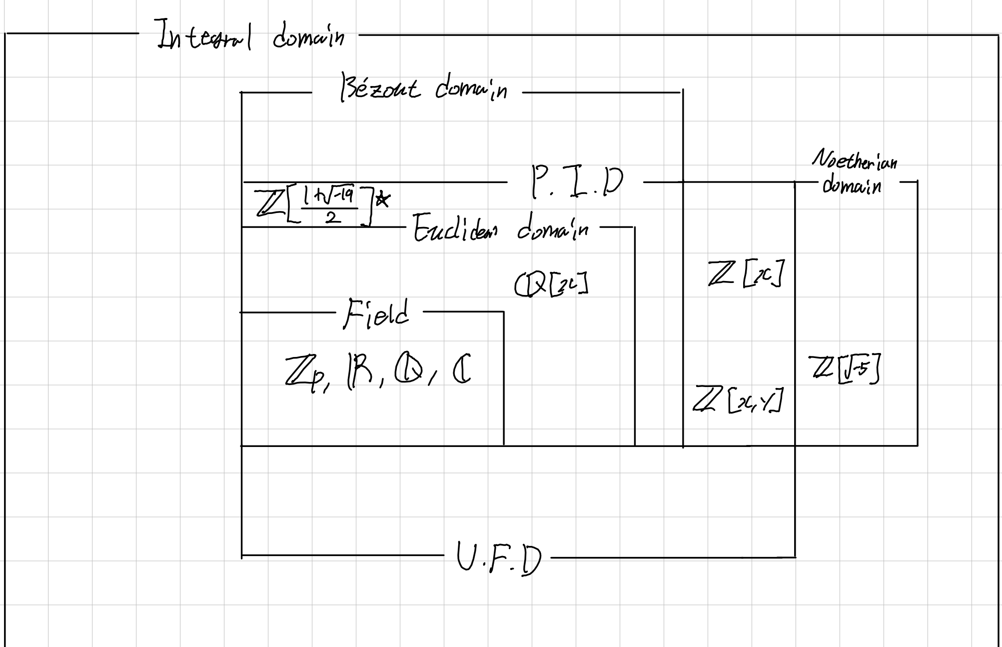
Field ⇒ Euclidean Domain ⇒ Principal Ideal Domain ⇒ Unique Factorization Domain ⊂ Integral Domain.

$$\text{Maximal} \underset{\text{I.D}}{\overset{\text{P.I.D}}{\Longleftrightarrow}} \text{Prime} \underset{\text{I.D}}{\overset{\text{P.I.D.}}{\Longleftrightarrow}} \text{Irreducible}.$$

$$\text{Maximal} \underset{\not\Leftarrow\ (2 \in \mathbb{Z}[x])}{\Rightarrow} \text{Prime} \overset{\text{U.F.D.}}{\Longleftrightarrow} \text{Irreducible}.$$

U.F.D

$$\text{P.I.D} \Rightarrow \begin{cases} \text{Noether} \Leftrightarrow \text{Every Ideal are finitely generated} \Leftrightarrow \text{A.C.C.I} \Rightarrow \text{A.C.C.P} \Rightarrow \exists\ \text{factorization} \\ \text{Bézout} \Leftrightarrow \text{Finitely generated ideal is principal} \Rightarrow \text{irreducible} = \text{Prime} \Rightarrow !\ \text{factorization} \end{cases} \Leftarrow \text{U.F.D}$$



「Commutative Ring with unity.

Basic Preliminaries; Let R be a Commutative Ring with unity $1 \neq 0$, $A \subseteq R$ is subset, $I \subseteq R$ Ideal.

1). $(A) = RA \overset{\text{def}}{=} \{r_1a_1 + \cdots + r_na_n \mid n \in \mathbb{N},\ r_i \in R,\ a_i \in A\}$

2). $I = R$ iff I has a unit. (not necessary Commutative)

3) R is field iff only Ideal of R is 0 or R.

4) $P \subsetneq R$ is prime $\Leftrightarrow$ R/P is integral domain.

5). $M \subsetneq R$ is Maximal $\Leftrightarrow$ R/M is field.

「Def. Let R be an Integral domain. A map

$$N : R \to \mathbb{N} \cup \{0\}$$

is called a Norm if $N(0) = 0$. If $N(a) > 0$ for all $a \neq 0$, then it is called positive Norm.

「Def. Let R be a Commutative Ring, and $a, b \in R$ with $b \neq 0$.

1). a is called a multiple of b if there exists $x \in R$ s.t. $a = bx$. Simply, denote $b \mid a$.

2). A non-zero $d \in R$ is called a greatest common divisor of a and b if

$$\begin{cases} d \mid a \text{ and } d \mid b \\ d' \mid a \text{ and } d' \mid b, \text{ then } d' \mid d. \end{cases}$$

Since $b \mid a \Leftrightarrow a \in (b) \Leftrightarrow (a) \subseteq (b)$, d is a greatest Common divisor of a and b iff

$$(a,b) \subseteq (d) \quad \text{and} \quad (a,b) \subseteq (d') \Rightarrow (d) \subseteq (d').$$

Directly,

「Prop. $(a,b) = (d) \Rightarrow d$ is g.c.d of a and b.

But the Converse is false, In $\mathbb{Z}[x]$, 1 is g.c.d of 2 and x but $R = (1) \neq (2,x)$.

Def. The integral domain R is said to be a Euclidean domain if: there exists a Norm N s.t for any $a, b \in R$ with $b \neq 0$, there exists $q, r \in R$ s.t $a = q \cdot b + r$ where $N(r) < N(b)$ or $r = 0$.

Lemma. If R is an Euclidean domain with a Norm N, then for any $a, b \in R$ with $b \neq 0$, there exists a sequence $\{r_k\}_{k=0}^{\infty}$ satisfying: put $r_{-2} = a$, $r_{-1} = b$,

$$\begin{cases} \text{for } 0 \leq k \leq n+1, \ r_{k-2} = q_k \cdot r_{k-1} + r_k \quad \text{for some } q_k \in R, \\ N(b) > N(r_0) > N(r_1) > \cdots > N(r_n), \\ r_{n+1} = r_{n+2} = \cdots = 0. \end{cases}$$

This $\{r_k\}_{k=0}^{\infty}$ is called reminder sequence for a and b.

Prop. Let R be an Euclidean domain. Then, for any non-zero ideal $I \subseteq R$,

$$I = (d), \text{ where } d \text{ is any non-zero element in } I \text{ of minimum norm.}$$

Proof. Let $I \subseteq R$ be a non-zero Ideal. observe that; the set

$$N[I \setminus \{0\}] = \{N(a) \mid a \in I, a \neq 0\}$$

has a minimum element $m \in \mathbb{N} \cup \{0\}$ by the Well-ordering principle, take $d \in N^{-1}[\{m\}]$. Trivially, $(d) \subseteq I$. To show the reverse inclusion, let $a \in I$ be given. Since R possess a Division Algorithm, for some $q, r \in R$,

$$a = qd + r \quad N(r) < N(d) \text{ or } r = 0$$

But, by the Minimality of $N(d)$, r must be zero. Therefore, $a = qd \in (d)$. ∎

Lemma. Let R be an Integral domain, let $d, d' \in R$.
If $(d) = (d')$, then $d = ud'$ for some unit $u \in R$.

Proof. If either d or d' is zero, there is nothing to prove.
Assume both d and d' are non-zero. Since $d \in (d')$ and $d' \in (d)$,

$$d = xd' \text{ and } d' = yd \text{ for some } x, y \in R.$$

Now, $d = xd' = xyd$ implies $d(1 - xy) = 0$.

Since R has no zero divisor and $d \neq 0$, $xy = 1$, x and y both are unit. ∎

Thm. Let R be an Euclidean domain, and a and b be non-zero elemenc in R.
Let $\{r_i\}$ be a remider sequence for a and b. Then,

$$(r_n) = (a, b).$$

Proof. Clearly, $r_n | r_{n-1}$, since $r_{n-1} = q_{n+1} r_n$. and $r_n | r_n$.
If r_n divides r_{k+1} and r_k, then r_n divides $r_{k-1} = q_{k+1} r_k + r_{k+1}$,
thus r_n divides r_k and r_{k-1}. Inductively, r_n divides $b = r_{-1}$ and $a = r_{-2}$
This proved that $(a, b) \subseteq (r_n)$. To show the reverse inclusion,
$a = q_0 b + r_0$ implies $r_0 \in (a, b)$ and $b = q_1 r_0 + r_1$ implies $r_1 \in (b, r_0) \subseteq (a, b)$
And, if $r_{k-1}, r_k \in (a, b)$, then $r_{k-1} = q_{k+1} r_k + r_{k+1}$ implies $r_{k+1} \in (a, b)$.
That is, $r_k, r_{k+1} \in (a, b)$. Inductively, $r_n \in (a, b)$. ∎

Def. An Integral domain R is called Principal Ideal Domain if:

Every Ideal in R is principal.

Directly, Euclidean domain $\Rightarrow$ P.I.D.

Prop. Let R be a P.I.D., and let a and b be non-zero elements of R.
Let $d \in R$ s.t $(d) = (a, b)$. Then,
1) d is a g.c.d. of a and b
2) $d = ax + by$ for some $x, y \in R$
3) d is unique up to multiplication by a unit of R.

Prop. Every nonzero prime ideal in a P.I.D is a Maximal.
Proof. Let $(p) \subsetneq R$ be a Prime Ideal, $p \neq 0$. And, let $m \in R$ s.t

$$(p) \subseteq (m) \subseteq R.$$

Enough to show that $(m) = (p)$ or $(m) = R$. By the assumption, $p = rm$ for some $r \in R$.
And, since (p) is prime, $rm \in (p)$ implies $r \in (p)$ or $m \in (p)$.
If $r \in (p)$, then $r = ps$ for some $s \in R$, $p = rm = psm$. Thus $sm = 1$, m is unit, $(m) = R$.
If $m \in (p)$, then $(m) \subseteq (p)$, thus $(m) = (p)$. ∎

The Converse is true in general integral domain.

Def. Let R be an Integral domain, let $r \in R$ be a nonzero, not a unit.

1). A nonzero, not unit $r \in R$ is called irreducible in R if:

if $r = ab$ with $a, b \in R$, then either a or b is a unit in R.

2). A nonzero $p \in R$ is called prime in R if: $(p) \subseteq R$ is prime ideal.

Equivalently, $p \in R$ is prime $\Leftrightarrow$ p is nonzero, not unit, $p \mid ab \Rightarrow p \mid a$ or $p \mid b$.

Lemma. Let R be an integral domain, and $r \in R$ be an element. Then,

r is irreducible element if and only if there is no element $m \in R$ s.t. $(r) \subsetneq (m) \subsetneq R$.

Proof. r is irreducible $\Rightarrow$ If $r = mx$, then either m or x is unit

$\Rightarrow (m) = R$ or $(m) = (r)$

r is reducible $\Rightarrow r = m \cdot n$ for some nonzero, non unit $m, n \in R \Rightarrow (r) \underset{\substack{\uparrow \\ n \text{ non unit}}}{\subsetneq} (m) \underset{\substack{\uparrow \\ m \text{ non unit}}}{\subsetneq} R$.

Therefore, $r \in R$ is irreducible $\Leftrightarrow$ (r) is maximal principal ideal.

(But, it is not necessarily (r) is maximal ideal.)

Prop. Prime element is irreducible in an integral domain.

Proof. Let $p \in R$ be a prime element in an integral domain.

If $p = ab$, then $p = ab \mid ab$, thus $p \mid a$ or $p \mid b$. Either $p \mid a$ or $p \mid b$,

$$\begin{cases} p = prb & \text{if } p \mid a \\ p = asp & \text{if } p \mid b \end{cases}$$

for some $r, s \in R$, which implies $rb = 1$ or $as = 1$, b or a is a unit. ∎

Prop. Irreducible element in P.I.D is a prime.

proof. Let $r \in R$ be an irreducible element in a P.I.D.

Suppose that an Ideal $M \subseteq R$ satisfies $(r) \subseteq M \subseteq R$. Since R is P.I.D, let $M = (m)$.

Since $r \in (m)$, $r = mx$ for some $x \in R$. And, r is irreducible, thus either m or x is a unit.

If m is an unit, then $(m) = R$, if x is an unit, then $m = rx^{-1} \in (r)$. Thus $(r) = (m)$.

Therefore, (r) is Maximal, thus prime.

In Sum, in P.I.D., Maximal $\xLeftrightarrow{\text{P.I.D}}$ Prime $\xLeftrightarrow{\text{P.I.D.}}$ Irreducible.

「More observation for principal Ideal.

Let R be a Ring, Not necessary identity, Commutative, no zero divisor.

Generally, if $I, J \subseteq R$ are Ideal, then $I + J = (I, J)$, the smallest ideal containing I and J.」

And, if $I, J, K \subseteq R$ are Ideals in a Ring R with identity 1, General Ring

$$I(J+K) = IJ + IK \quad \text{and} \quad (I+J)K = IK + JK$$

Proof. Let $x \in I(J+K)$. Then $x = a_1 b_1 + \cdots + a_n b_n$, $a_i \in I$, $b_i \in J+K$

$= a_1(j_1 + k_1) + \cdots + a_n(j_n + k_n)$, $j_n \in J$, $k_n \in K$

$= a_1 j_1 + \cdots + a_n j_n + a_1 k_1 + \cdots + a_n k_n \in IJ + JK$.

And, since $J, K \subseteq J+K$, $\underset{\text{Need identity?}}{IJ, IK \subseteq I(J+K)}$, thus $IJ + IK \subseteq I(J+K)$ ∎ Ring with 1.

This allows to us that: for any $a, b, c \in R$ where R contains 1,

$$(a, b) = (a) + (b)$$

$$(a)[(b) + (c)] = (a)(b) + (a)(c) = (ab) + (ac)$$

$$[(a) + (b)](c) = (a)(c) + (b)(c) = (ac) + (bc)$$

「Thm. Let R be a Ring, and $I, J \subset R$ be Ideals. Then,

$$(I, J) \overset{\text{def}}{=} \bigcap_{\substack{S \text{ Ideal} \\ I \cup J \subseteq S}} S = I + J.$$

Proof. 1). The sum of ideals is Ideal.

Let $a + b \in I + J$ be given, and let $r \in R$. Then,

$$r \cdot (a+b) = ra + rb \in I + J \quad \text{and} \quad (a+b) \cdot r = a \cdot r + b \cdot r \in I + J.$$

And, for $a_1, a_2 \in I$ and $b_1, b_2 \in J$, $\in I$ $\in J$, by Ideal

$$(a_1 + b_1) \cdot (a_2 + b_2) = \underbrace{a_1 a_2}_{\in I} + a_1 b_2 + b_1 a_2 + \underbrace{b_1 b_2}_{\in J} \in I + J.$$

by I, J are Subring

And, $I, J \subseteq I + J$ clearly, so $(I, J) \subseteq I + J$. And, $a \in I$, $b \in J \Rightarrow a + b \in (I, J)$.

Def. A Unique Factorization Domain is an integral domain R in which every nonzero, not unit $r \in R$ satisfies the following two properties:

1). r can be written as a finite product of irreducibles $p_i \in R$. (Not necessary distinct)

2). This representation is unique up to associates.

Prop. In a U.F.D., prime <=> irreducible.

Proof. Generally, prime element is irreducible. Let $p \in R$ be an irreducible in a U.F.D R.
Assume that $p \mid ab$ for some $a, b \in R$. That is, $ab = pc$ for some $c \in R$.
Since R is U.F.D. and a, b are nonzero, not unit, a and b have factorizations;
(If either a or b is unit, then p also must divides other one)

$$a = u \cdot p_1 \cdots p_n, \quad b = v \cdot q_1 \cdots q_m \quad (u, v \text{ are unit, } p_i, q_j \text{ are irreducible})$$

Now,
$$ab = uv\, p_1 \cdots p_n q_1 \cdots q_m = pc$$

and this factorization is unique up to associate, p must associate to one of $p_1, \cdots, p_n, q_1, \cdots, q_m$.
Thus, $p \mid a$ or $p \mid b$. ∎

In Sum, in U.F.D, $\text{Maximal} \overset{?}{\Longleftrightarrow} \text{Prime} \overset{\text{U.F.D.}}{\Longleftrightarrow} \text{Irreducible}$.

⌈Thm. Every Principal Ideal Domain is Unique Factorization Domain.

Proof. Let R be a P.I.D., and $r \in R$ be a nonzero, not unit.
If r is irreducible, then clear. Suppose r is reducible. Then,
$r = r_1 r_2$ for some non unit r_1, r_2. If both r_1 and r_2 are irreducible, clear.
If either r_1 or r_2 is not irreducible, WLOG, r_1 is reducible,
$r_1 = r_{11} r_{12}$ for some nonunit r_{11}, r_{12}. Clearly,

$$(r) \subsetneq (r_1) \subsetneq (r_{11}) \subsetneq R$$

because $r \in (r_1)$ since $r = r_1 r_2$, but $r_1 \notin (r)$, since r_2 is not unit.
(specifically, if $r_1 \in (r)$, then $r_1 = rx = r_1 r_2 x \Rightarrow r_1(1 - r_2 x) = 0 \Rightarrow r_2 x = 1 \Rightarrow r_2$ unit)
Inductivelly, we can Construct an Ascending chain of ideals:

$$(r) \subsetneq (r_1) \subsetneq (r_{11}) \subsetneq \cdots \subsetneq R$$

But, Every Ascending Chain in P.I.D will be stationary. Because:
Let $\{I_n\}_{n=1}^{\infty}$ be a Collection of Ideals of P.I.D R s.t.

$$I_1 \subseteq I_2 \subseteq \cdots \subseteq R.$$

Denote $I = \bigcup_{n=1}^{\infty} I_n$. Then, it is Ideal, because for any $r \in R$ and $a \in I$, $a \in I_n$ for some $n \in \mathbb{N}$,
thus $ra \in I_n \subseteq I$. Also, I is a subring, because $\{I_n\}$ is nested. Thus I is ideal.
Since R is P.I.D., denote $I = (d)$ for some $d \in R$. Since $d \in I$, $d \in I_n$ for some $n \in \mathbb{N}$.
This means $I_n \subseteq I = (d) \subseteq I_n$, thus $I_n = I$. ⌋ P.I.D $\Rightarrow$ Noether Domain.

Therefore there is No infinite strictly ascending chain,
thus r has a finite irreducible factorization. Now the Uniqueness remains.
Using induction for number of irreducible factors for some factorization of r.
If $n=1$, then r is irreducible. If $r = p_1 p_2 \cdots p_m$ is other factorization where p_i are irreducible,
then r must divides one of p_i, because r is prime. That is, $p_i = u \cdot r$ for some i.
WLOG, put $i=1$. Then, $r = u r p_2 \cdots p_m \Rightarrow r(1 - u p_2 \cdots p_m) = 0$, but since $p_2, \cdots, p_m$ are not unit,
it can not possible. Suppose for $n-1$, the Uniqueness holds. If r has two factorization

$$r = p_1 p_2 \cdots p_n = q_1 q_2 \cdots q_m \qquad m \geq n$$

Similarly, p_1 must divide one of q_i, WLOG q_1. Then $p_2 \cdots p_n = u q_2 \cdots q_m$, for some unit u,
These are same representation, by the induction.

Def. An integral domain R in which every ideal generated by two elements is principal is called a Bézout Domain. Simply, for all $a, b \in R$, there exists $d \in R$ s.t.

$$(a,b) = (a) + (b) = (d)$$

Prop. Let R be an integral domain.

R is Bézout Domain if and only if for any $a, b \in R$, $d = \gcd(a,b)$ exists with $d = ax + by$ for some $x, y \in R$.

Proof. Suppose R is Bézout domain. Let $a, b \in R$ be given. Then,

$$(a,b) = (d) \text{ for some } d \in R.$$

Now, $d = ax + by$ for some $x, y \in R$, and, since $(a), (b) \subseteq (a,b) = (d)$, $d \mid a$ and $d \mid b$. And, if $d' \mid a$ and $d' \mid b$, then $d' \mid d (= ax + by)$. The converse is trivial.

Thm. Every finitely generated Ideal of a Bézout domain is principal.

Proof. Let R be a Bézout domain, and $I \subseteq R$ be a finitely generated Ideal. Denote

$$I = (a_1, \cdots, a_n) \text{ for some } a_1, \cdots, a_n \in R.$$

Using induction for n. If $n = 1$ or 2, trivial. Suppose for $n-1$ the statement holds.
Then, $(a_1, \cdots, a_{n-1}) = (p)$ for some $p \in R$. Enough to show $(a_1, \cdots, a_n) = (p, a_n)$.

Let $a_1x_1 + \cdots + a_nx_n \in (a_1, \cdots, a_n)$ be given. Since for each a_i, $p \mid a_i$, thus

$$a_1x_1 + \cdots + a_{n-1}x_{n-1} + a_nx_n = px + a_nx_n \in (p, a_n).$$

Converse is trivial. ▨ (Simply, $(p, a_n) = (p) + (a_n) = (a_1, \cdots, a_{n-1}) + (a_n) = (a_1, \cdots, a_n)$.)

Thm. If R is Bézout domain and also U.F.D., then it is P.I.D.

Proof. Let $I \subseteq R$ be an Ideal of R, and $a \in I$ be a nonzero, minimal number of irr. factors.
If $I \neq (a)$, then there is $b \in I \setminus (a)$. Since R is Bézout domain, for some $d \in I$,

$$(d) = (a,b) \subseteq I.$$

That is, $d \mid a$ and $d \mid b$. But, since the minimality of a, d must associates to a.
But, this implies $d \mid b \Rightarrow b \in (d) = (a)$, Contradiction. ▨

Prop. In Bézout domain, every irreducible element generates Maximal Ideal, thus prime.

Proof. Let $r \in R$ be an irreducible. Let M be an Ideal in R s.t $(r) \subseteq M \subseteq R$.
Take $x \in M \setminus (r)$. [If no such x exists, then $M = (r)$, thus (r) is maximal $\Rightarrow$ prime.]
Clearly, $(r) \subsetneq (r, x) \subseteq M$. Since R is a Bézout domain, there exists $d \in R$ s.t

$$(r, x) = (r) + (x) = (d)$$

But, since r is irreducible, d must be an unit. That is, $(r, x) = R$
Consequently, (r) is Maximal, thus Prime ideal.

+ More Precisely, $d \mid r \Leftrightarrow r = ds$ for some $s \in R$. Since r is irreducible, either d or s is an unit. If s is a unit, then

$$d = rs^{-1} \mid x \quad \Rightarrow \quad x = rs^{-1}t \in (r) \text{ for some } t \in R$$

But, it is Contradiction. Therefore d must be an unit.

Def. An Integral Domain N is called a Noether Domain if:

If a sequence of Ideals $\{I_n\}$ in N satisfies that:

there are no infinite increasing chains of Ideals.

Thm. Let R be an Integral domain.

R is noether domain if and only if Every Ideal of R is finitely generated.

proof. Suppose that R is noether domain. Assume R has an ideal I, which cannot be generated finitely. Then, we can construct $\{I_n\}$ as follow:

given $a_1 \in I$, $I_1 = (a_1) \subseteq I$

given $a_2 \in I \setminus I_1$, $I_2 = (a_1, a_2) \subseteq I$

given $a_3 \in I \setminus I_2$, $I_3 = (a_1, a_2, a_3) \subseteq I$

$\vdots$

Then, $I_1 \subsetneq I_2 \subsetneq \cdots \subsetneq I$, because $a_n \notin I_{n-1}$, so $I_{n-1} \neq I_n$

But, this is contradiction to R satisfies ACC on Ideals.

Conversely, let $\{I_n\}_{n=1}^{\infty}$ be a chain of ideals in R s.t

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$$

Take $I = \bigcup_{n=1}^{\infty} I_n$. Then, for any $a, b \in I$, $a, b \in I_N$ for some $N \in \mathbb{N}$.

Since I_N is a subring, $a-b \in I_N \subseteq I$ and $ab \in I_N \subseteq I$, so I is subring, morover, for any $r \in R$, $ra \in I_N \subseteq I$, so it is an Ideal.

By the hypothesis, I is generated finitely. That is,

$$I = (a_1, \cdots, a_n) \text{ for some } a_1, \cdots, a_n \in I.$$

Since each $a_i \in I$, there is $N_i \in \mathbb{N}$ s.t $a_i \in I_{N_i}$. Now, take $m = \max\{N_1, \cdots, N_n\}$,

$$I = (a_1, \cdots, a_n) \subseteq I_m \subseteq I$$

■

Note: For a field F, $F[x_1, x_2, \cdots]$ is Not noether domain:

$$(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \cdots$$

But, $F[x_1, x_2]$ is Noetherian, by the Hilbert basis Theorem.

Since $F[x_1, x_2]$ is Not P.I.D., it is also Not Bezout.

「$F[x,y]$.

Let F be a field, and let x, y be independent indeterminates.

1. $F[x,y]$ is a Noetherian Domain.

Proof. Since $F[x,y] = (F[x])[y]$, the Hilbert basis Theorem gives that the result

2. $F[x,y]$ is <u>Not</u> P.I.D.

Proof. Consider the ideal (x,y). If (x,y) is principal then

$$(x,y) = (a(x,y)) \text{ for some } a(x,y) \in F[x,y]$$

That is, $x = f(x,y) \cdot a(x,y)$, for some $f(x,y) \in F[x,y]$.

If $a(x,y)$ is constant, then (x,y) contains a unit, so it is entire. Contradiction.

So, $a(x,y)$ must be $a \cdot x$, where a is non-zero constant.

Similarly, $a(x,y) = b \cdot y$ for some non-zero $b \in F$, $ax = by$, Contradiction.

3. $F[x,y]$ is <u>Not</u> Bezout domain.

4. x and y in $F[x,y]$ are relatively prime, but $F[x,y] = (1) \neq (x) + (y)$.

⌈Example. Let $R = \mathbb{Z}[x]$. Then, $(2,x) \subset R$ has the following properties:

$$(2,x) = \{2p(x) + x\,q(x) \mid p, q \in \mathbb{Z}[x]\} \subsetneq R, \text{ since } 1 \notin (2,x). \text{ Thus proper.}$$

If $(2,x) = (a(x))$ for some $a(x) \in \mathbb{Z}[x]$, then, $2 \in (a(x))$, this means

$$2 = a(x) \cdot p(x) \text{ for some } p(x) \in \mathbb{Z}[x].$$

Now, $a(x), p(x) \in \{\pm 1, \pm 2\}$ If $a(x) = \pm 1$, then $(a(x)) = (1) = R$, Contradiction.
thus $a(x) = \pm 2$. But, $x \notin (\pm 2)$, Contradiction. Now, $(2,x)$ is not principal.
But $(2,x) \subsetneq \mathbb{Z}[x]$ is Maximal, because a map

$$\varphi : \mathbb{Z}[x] \to \mathbb{Z}_2 : p(x) \mapsto p(0) \bmod 2 \qquad \mathbb{Z}[x] \to \mathbb{Z} \to \mathbb{Z}_2.$$

is an onto Ring-Homomorphism having $\ker \varphi = (2,x)$ because

$$p(x) \in \ker \varphi \iff p(0) \bmod 2 = 0 \iff p(x) \in (2,x)$$

Thus $\mathbb{Z}[x]/\ker\varphi \cong \mathbb{Z}_2$ is a field, $\ker \varphi = (2,x)$ a Maximal, further, Prime.
And, clearly, $2 \in R$ is irreducible but $(2) \subsetneq (2,x)$, it is Prime, but not Maximal.

$$(x) = \{p(x) \in \mathbb{Z}[x] \mid p(0) = 0\}.$$

Note: $(2,x) \subseteq \mathbb{Z}[x]$ is also maximal, because:
$\mathbb{Z}[x]/(2) = \mathbb{Z}[x]/2\mathbb{Z}[x] \cong (\mathbb{Z}/2\mathbb{Z})[x]$, with the natural Homomorphism

$$\pi : \mathbb{Z}[x] \to (\mathbb{Z}/2\mathbb{Z})[x] : \sum a_n x^n \mapsto \sum (a_n \bmod 2) \cdot x^n$$

And consider the evaluation Homomorphism

$$\varphi : (\mathbb{Z}/2\mathbb{Z})[x] \to \mathbb{Z}/2\mathbb{Z} : p(x) \mapsto p(0)$$

Then, $\ker \pi = 2\mathbb{Z}[x] = (2)$ and $\ker \varphi = \{p(x) \in (\mathbb{Z}/2\mathbb{Z})[x] \mid p(0) = 0\} = (x)$
Moreover, since π and φ are surjective, so $\varphi \circ \pi$ is also surjective,

$$\ker \varphi \circ \pi = \{p(x) \in \mathbb{Z}[x] \mid p(0) \bmod 2 = 0\} = (2, x)$$

$p(x) = p(0) + x \cdot q(x)$ where $q(x) = (p(x) - p(0))/x$ or 0.
So, $p(0) \equiv 0 \bmod 2$
$\Rightarrow p(x) = 2 \cdot k + x \cdot q(x) \in (2) + (x)$

⌈Example. The polynomial ring $\mathbb{Z}[x]$ is <u>Not</u> P.I.D., but U.F.D.
Specifically, $(2,x) \subseteq \mathbb{Z}[x]$ is Not Principal, but Maximal.
And, $(2) \subseteq \mathbb{Z}[x]$ is Irreducible and Prime, but <u>Not</u> Maximal.

⌈Example. The quadratic integer ring $\mathbb{Z}[\sqrt{-5}]$ is <u>Not</u> U.F.D., but Noether Domain.
Not U.F.D: $6 = 2 \cdot 3 = (1+\sqrt{-5}) \cdot (1-\sqrt{-5})$ and
$2, 3, 1 \pm \sqrt{-5}$ are irreducible

「Quadratic Integer Ring with Norm.

Def. Let D be a square free integer. Define

$$\mathbb{Z}[\sqrt{D}] \overset{\text{def}}{=} \{a + b\sqrt{D} \mid a, b \in \mathbb{Z}\}$$

is called a Quadratic Integer Ring. And, define a Norm on $\mathbb{C}$ as

$$N : \mathbb{C} \to \mathbb{R} ; a + i \cdot b \mapsto a^2 + b^2$$

or simply, $N(\alpha) := \alpha \cdot \bar{\alpha}$.

「Prop. $N(\alpha) = 0$ if and only if $\alpha = 0$

Proof. $N(\alpha) = 0 \Leftrightarrow a^2 + b^2 = 0$ where $\alpha = a + ib$. Thus clear.

「Prop. $N(\alpha\beta) = N(\alpha)N(\beta)$

Proof. Denote $\alpha = a + ib$, $\beta = c + id$.

$$\begin{aligned} N(\alpha\beta) &= N(ac - bd + i(ad + bc)) \\ &= (ac - bd)^2 + (ad + bc)^2 \\ &= a^2c^2 + b^2d^2 - 2abcd + a^2d^2 + b^2c^2 + 2abcd \\ &= (a^2 + b^2)(c^2 + d^2) = N(\alpha)N(\beta) \end{aligned}$$

Example. The Quadratic Integer $\mathbb{Z}[\sqrt{-5}]$.

1). $\mathbb{Z}[\sqrt{-5}]$ has no zero divisor. (Thus Integral domain)

Proof. If $\alpha, \beta \in \mathbb{Z}[\sqrt{-5}]$ satisfies $\alpha\beta = 0$, then

$$N(\alpha\beta) = N(\alpha)N(\beta) = 0$$

This implies $N(\alpha)=0$ or $N(\beta)=0$, that is, $\alpha=0$ or $\beta=0$.

2). $\mathbb{Z}[\sqrt{-5}]$ has units only ± 1.

Proof. Observe that $\alpha\beta = 1 \Rightarrow N(\alpha\beta) = N(\alpha)N(\beta) = 1$.

Since $N[\mathbb{Z}[\sqrt{-5}]] \subseteq \mathbb{N} \cup \{0\}$, $N(\alpha) = N(\beta) = 1$

But, $N(\alpha) = a^2 + 5b^2 = 1$ iff $a = \pm 1$ and $b = 0$.

Thus $\alpha = \beta = 1$ or $\alpha = \beta = -1$.

3). $2, 3, 1 \pm \sqrt{-5}$ are irreducible, and no two which are not associate.

Proof. If 2 is reducible, then $2 = \alpha\beta$ for some $\alpha, \beta \in \mathbb{Z}[\sqrt{-5}]$ s.t. $\alpha, \beta \neq \pm 1$.

But, $4 = N(2) = N(\alpha\beta) = N(\alpha)N(\beta)$ implies $N(\alpha)=1, N(\beta)=4$ or $N(\alpha)=N(\beta)=2$ or $N(\alpha)=4, N(\beta)=1$

Since $N(\alpha)=1$ iff $\alpha=\pm 1$, the only possible case is $N(\alpha)=N(\beta)=2$. But,

$a^2+5b^2=2$ has no Integer solution. Similarly, 3 is irreducible.

(But, 5 is reducible, since $5 = (\sqrt{-5}) \cdot (-\sqrt{-5})$)

And, if $1 \pm \sqrt{-5}$ is reducible, then $1 \pm \sqrt{-5} = \alpha\beta$ with $\alpha, \beta \neq \pm 1$. Then,

$$6 = N(1 \pm \sqrt{-5}) = N(\alpha\beta) = N(\alpha)N(\beta)$$

Then, since $N(\alpha), N(\beta) \neq 1$, $N(\alpha) = 2$, $N(\beta) = 3$ or $N(\alpha)=3$, $N(\beta)=2$.

But, $a^2+5b^2 = 2$ or 3 have no integer solution, thus $1 \pm \sqrt{-5}$ are irreducible.

And, not associated is clear, because $\mathbb{Z}[\sqrt{-5}]$ has unit only ± 1.

4). $\mathbb{Z}[\sqrt{-5}]$ is Not U.F.D.

$$6 = 2 \cdot 3 = (1+\sqrt{-5}) \cdot (1-\sqrt{-5}).$$

∎

5). $2 \in \mathbb{Z}[\sqrt{-5}]$ is irreducible but not prime.

$2 \mid 6 = 3 \cdot 2 = (1+\sqrt{-5})(1-\sqrt{-5})$, but $2 \nmid 1 \pm \sqrt{-5}$.

5). $\mathbb{Z}[\sqrt{-5}]$ is Noether domain. (Thus, not Bézout) $\mathbb{Z}[x,y]$

Define a Homomorphism;

$$\varphi;\ \mathbb{Z}[x] \to \mathbb{Z}[\sqrt{-5}] : P(x) \mapsto P(\sqrt{-5})$$

$$\underbrace{\mathbb{Z}[x]}_{\text{U.F.D.}} / (\underbrace{x^2+5}_{\substack{\text{irr.} \\ \Downarrow \\ \text{Prime}}}) \cong \mathbb{Z}[\sqrt{-5}]$$

$\mathbb{Z}$ E.D $\Rightarrow$ P.I.D $\Rightarrow$ Noether $\Rightarrow$ $\mathbb{Z}[x]$ noether $\Rightarrow$ $\mathbb{Z}[\sqrt{-5}]$ noether

6). $11 \in \mathbb{Z}[\sqrt{-5}]$ is Prime element, but (11) is <u>not</u> Maximal.

Let $a+b\sqrt{-5} \in \mathbb{Z}[\sqrt{-5}]$

7). $1+\sqrt{-5}$ is not prime,

observe that $(1+\sqrt{-5})(1-\sqrt{-5}) = 1-(-5) = 6 = 2\cdot 3$,

If $1+\sqrt{-5}$ divides 2, then $N(2) = 4 = N(1+\sqrt{-5})\cdot N(\alpha) = 6\cdot N(\alpha)$, there is no such α.

If $1+\sqrt{-5}$ divides 3, then $9 = 6\cdot N(\alpha)$, similarly, there is no such α.

「Example. The Quadratic Integer Ring $\mathbb{Z}[\sqrt{-5}]$

1). $\mathbb{Z}[\sqrt{-5}]$ has no zero divisor.

Suppose that $(a+b\sqrt{-5})(c+d\sqrt{-5}) = ac-5bd+(ad+bc)\sqrt{-5} = 0.$

It equals to; $\begin{cases} ac-5bd=0 \\ ad+bc=0 \end{cases}$. Then,

$adc = 5bd^2 \iff -bdc^2 = 5bd^2 \iff bd(5d-c^2)=0.$

Thus, $b=0$ or $d=0$ or $5d=c^2$.

$b=0 \Rightarrow ac=0 \Rightarrow \begin{cases} a=0 \Rightarrow a+b\sqrt{-5}=0. \\ c=0 \Rightarrow ad=0 \Rightarrow \begin{cases} a=0 \Rightarrow a+b\sqrt{-5}=0 \\ d=0 \Rightarrow c+d\sqrt{-5}=0. \end{cases} \end{cases}$

$d=0 \Rightarrow ac=0 \Rightarrow \begin{cases} a=0 \Rightarrow bc=0 \Rightarrow \begin{cases} b=0 \Rightarrow a+b\sqrt{-5}=0 \\ c=0 \Rightarrow c+d\sqrt{-5}=0. \end{cases} \\ c=0 \Rightarrow c+d\sqrt{-5}=0 \end{cases}$

$5d=c^2 \Rightarrow ac-5bd = ac-dc^2 = c(a-dc)=0$

$\Rightarrow \begin{cases} c=0 \Rightarrow c+d\sqrt{-5}=0 \\ a=dc \Rightarrow adc+bc^2 = a^2+5bd=0 \end{cases}$

1). Only unit of $\mathbb{Z}[\sqrt{-5}]$ is ± 1.

Suppose that $(a+b\sqrt{-5})(c+d\sqrt{-5}) = ac-5bd+(ad+bc)\sqrt{-5} = 1.$

It equals to; $\begin{cases} ac-5bd=1 \\ ad+bc=0 \end{cases}$.

Fail Try

「Fail try

And, let $P \subsetneq R$ be a Prime ideal. Let M be an Ideal s.t $P \subseteq M \subseteq R$.

$$(ab) \subseteq (a), (b)$$
$$\subseteq (a) + (b)$$

$$(a,b) = (a) + (b) = (d)$$

To show r is prime, we will show that (r) is maximal.

Suppose that $r | ab$ for some $a, b \in R$.

That is, $ab = rx$ for some $x \in R$.

Since R is Bézout domain, there exists $d \in R$ s.t $(d) = (a,b)$

Suppose that $r | ab$. That is, $ab \in (r)$.

Since R is Bézout domain, $(a) + (b) = (d)$ for some $d \in R$, and $(ab) \subseteq (a), (b)$ thus

$$(ab) \subseteq (a) + (b) = (d)$$

But, since d is greatest common divisor of a and b,

$$(r) \subsetneq (d)$$

$r | a$ and $r | b$ $\quad d = r$

$d | ab$